HW3: Simple Structure from Motion

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COMS4732W Computer Vision 2

Overview

This project implements components of a simple Structure from Motion (SfM) pipeline that estimates correspondences, camera pose, and 3D scene points from a pair of images:

- **Step 1:** Camera Intrinsics (Calculated from iPhone 15 Pro specifications)
- **Step 2:** Detecting SIFT Features (rotation + scale invariant)
- **Step 3:** Feature Matching using L2 distance and Nearest Neighbor Distance Ratio (NNDR)
- **Step 4:** RANSAC to estimate Rotation (R) and Translation (t) via Essential Matrix decomposition
- **Step 5:** Triangulating inliers to generate a 3D Sparse Point Cloud

Selected Configuration Parameters

Parameter	Value
Random Seed	`42`

Camera Intrinsics Matrix (K)	`[[ 887.47, 0.00, 640.00 ],`
	`[ 0.00, 887.47, 480.00 ],`
	`[ 0.00, 0.00, 1.00 ]]`

SIFT Points Detected	`1197` features in both images (max capped at 1200)
Feature Matching NNDR Threshold	`0.8`
RANSAC Number of Iterations	`2000`

RANSAC Epsilon (Sampson distance)	`0.01`

Scene 1: Bookshelf Room

▼ Visualizations — Pipeline Steps

▼ Overview: Full Pipeline Grid

Complete Pose Estimation Pipeline

Image 1: Original

Image 2: Original

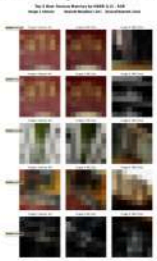
SIFT (1197)

SIFT (1197)

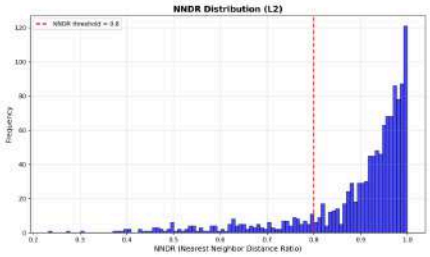
SIFT (No ANMS) (1197)

SIFT (No ANMS) (1197)

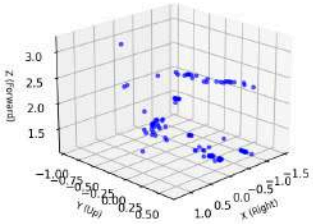
NNDR Top 5 Matches



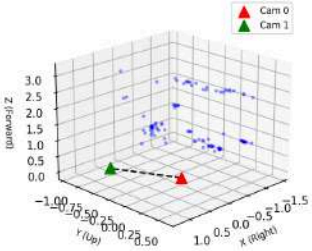
NNDR Ratio Histogram



3D Points (105)

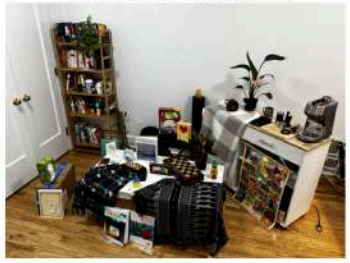


Camera Poses



Matches Before RANSAC (141)

Matches Before RANSAC (141)



```
FEATURE DETECTION

SIFT:
  Image 1: 1197
  Image 2: 1197

ANMS Selected:
  Image 1: 1197
  Image 2: 1197
Config:
  Max features: 1200
  Contrast thresh: 0.04
  Edge thresh: 10

FEATURE MATCHING

Before RANSAC:
  Correspondences: 141

After RANSAC:
  Inliers: 105
  Inlier Ratio: 74.5%
Config:
  Metric: L2
  NNDR threshold: 0.8
```

```
POSE ESTIMATION

Translation:
  X: -0.7689
  Y: 0.5328
  Z: 0.3536

Baseline: 1.0000
RANSAC Config:
  Sample size (s): 8
  Epsilon: 0.0002
  Iterations: 2000

3D RECONSTRUCTION

Method: Sparse
Points: 105
Mean depth: 1.73
Triangulation Config:
  Method: Sparse features
  Max reproj error: 50.0px
  Depth range: [0.81, 10000]
```

Inliers After RANSAC (105)

Inliers After RANSAC (105)

Image 1 - Feature Points

Image 2 - Epipolar Lines

Image 2 - Feature Points

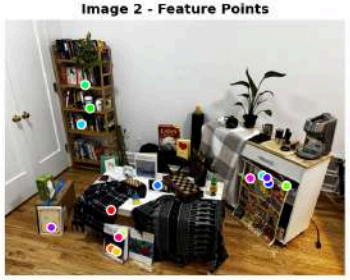
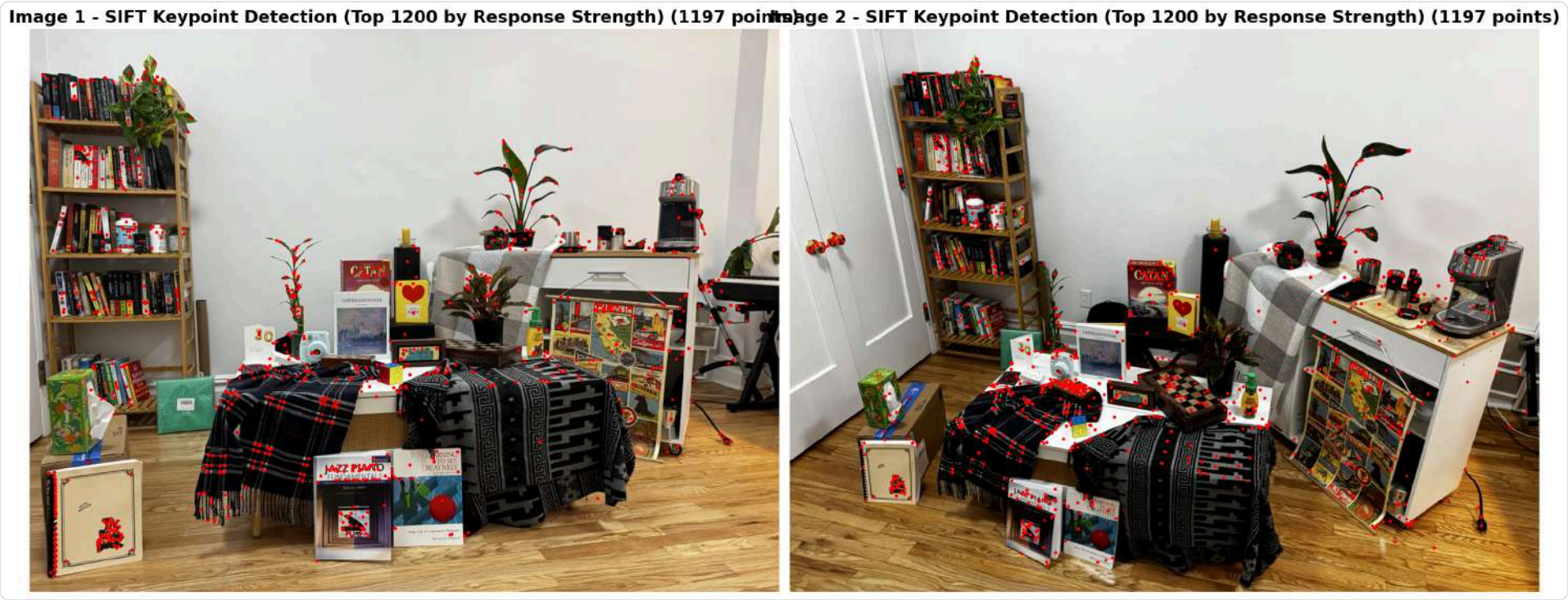
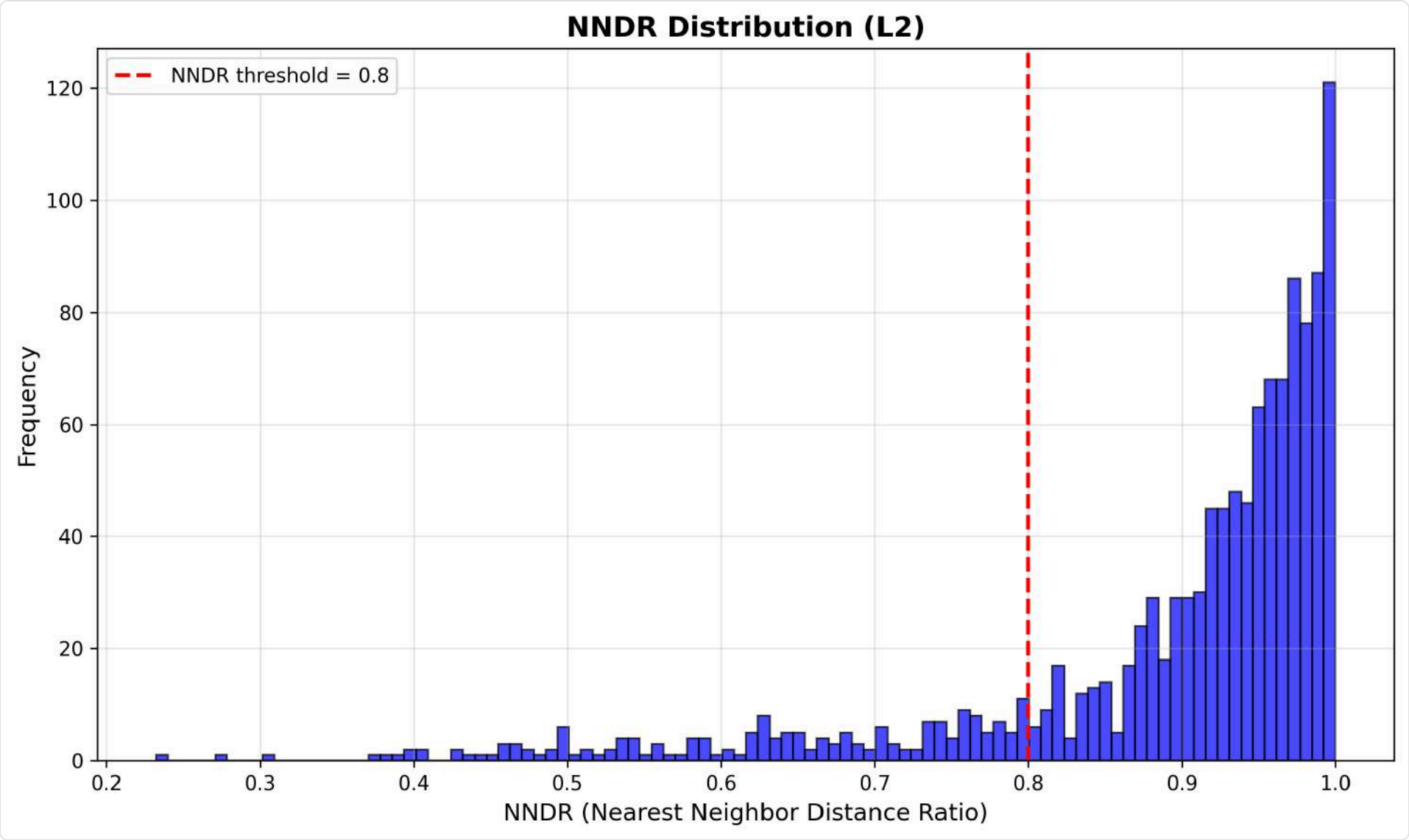


Image 1 - Epipolar Lines

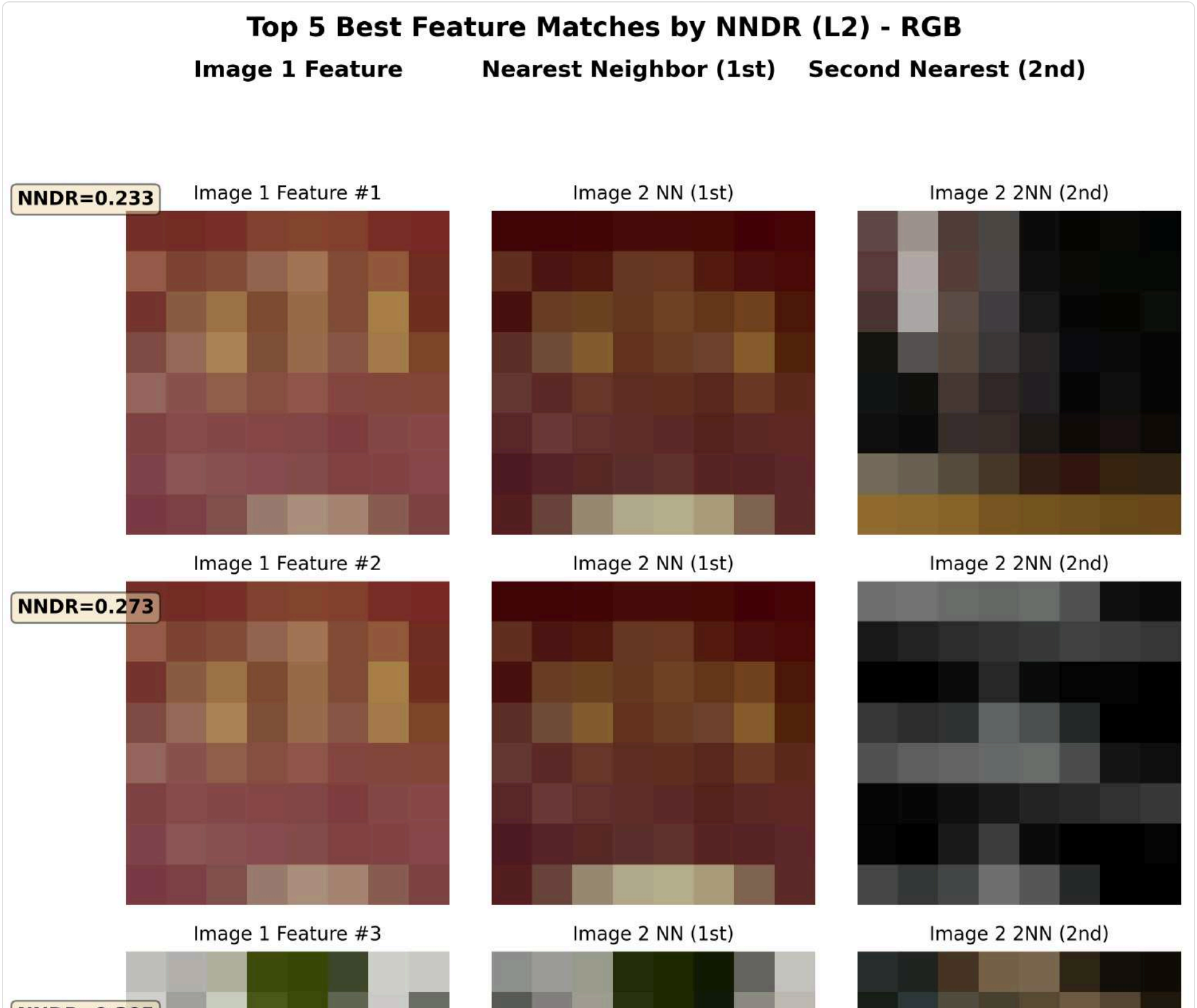
▼ Step 2: SIFT Features Detected



▼ Step 3: NNDR Histogram



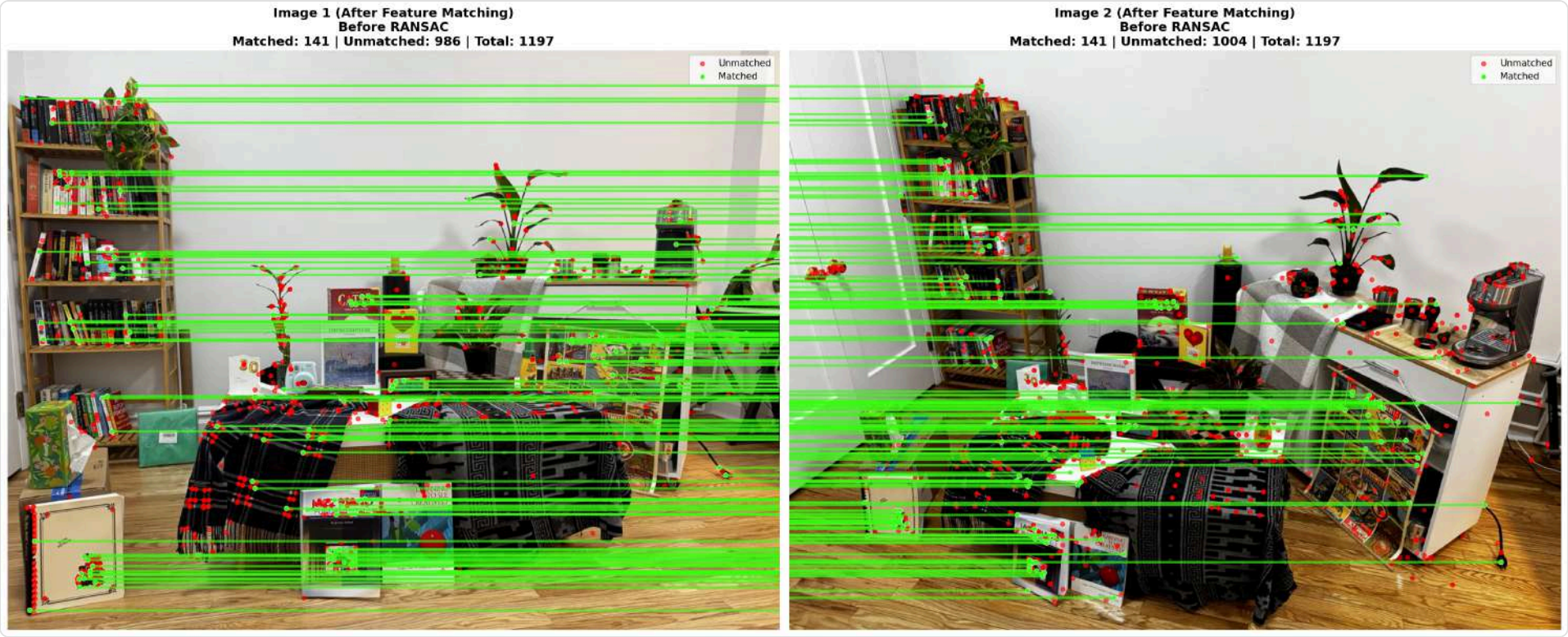
▼ Step 3: Top 5 Features Matched (Ranked by NNDR)


Top 5 Best Feature Matches by NNDR (L2) - RGB
Image 1 Feature
Nearest Neighbor (1st)
Second Nearest (2nd)
NNDR=0.233
Image 1 Feature #1
Image 2 NN (1st)
Image 2 2NN (2nd)
NNDR=0.273
Image 1 Feature #2
Image 2 NN (1st)
Image 2 2NN (2nd)
Image 1 Feature #3
Image 2 NN (1st)
Image 2 2NN (2nd)

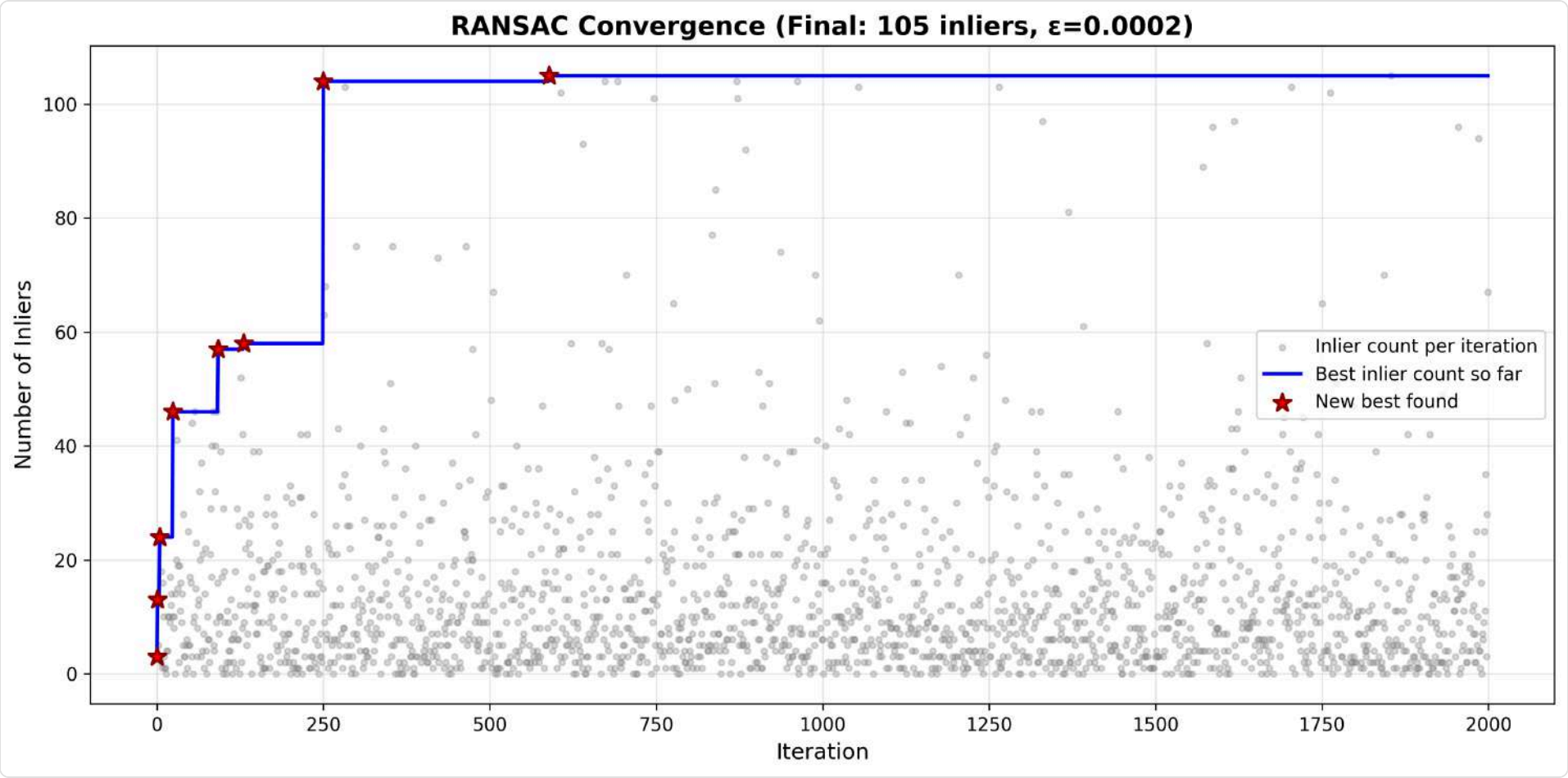
NNDR=0.305
Image 1 Feature #4
Image 2 NN (1st)
Image 2 2NN (2nd)
NNDR=0.372
Image 1 Feature #5
Image 2 NN (1st)
Image 2 2NN (2nd)
NNDR=0.386

▼ Step 3: Correspondences Before RANSAC

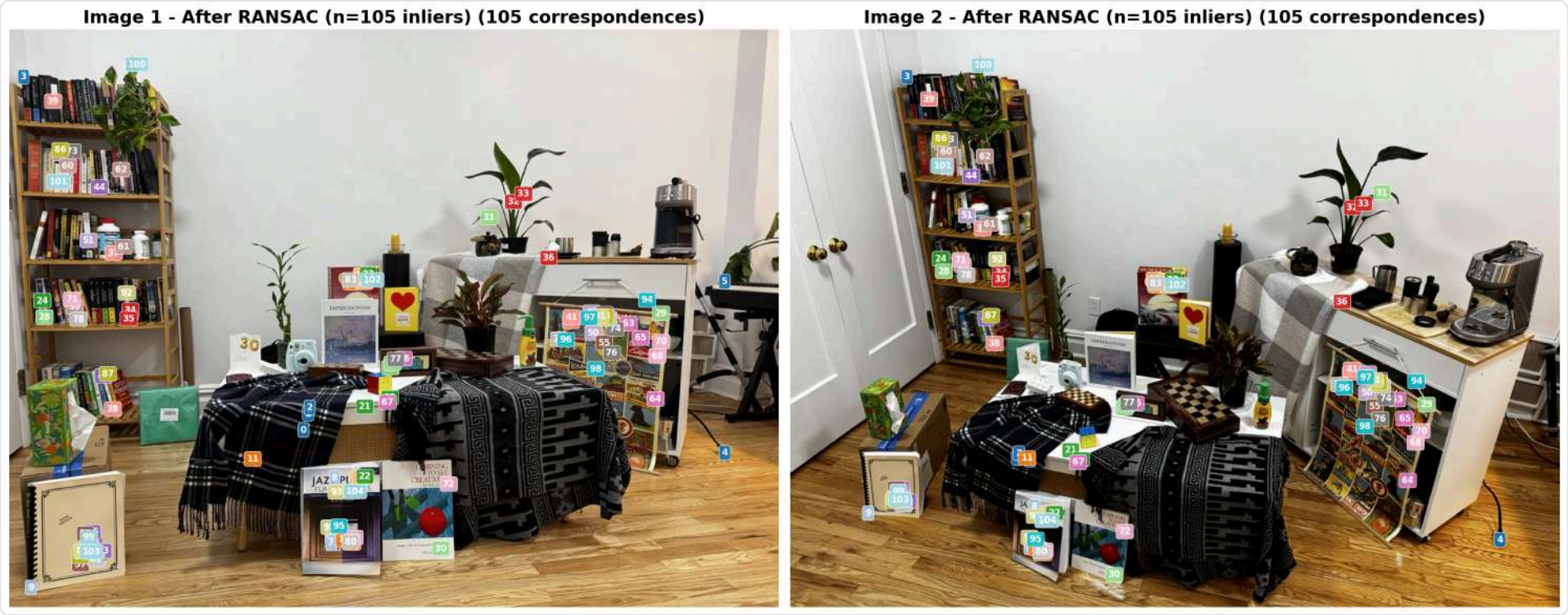
Alternate view (with patches):



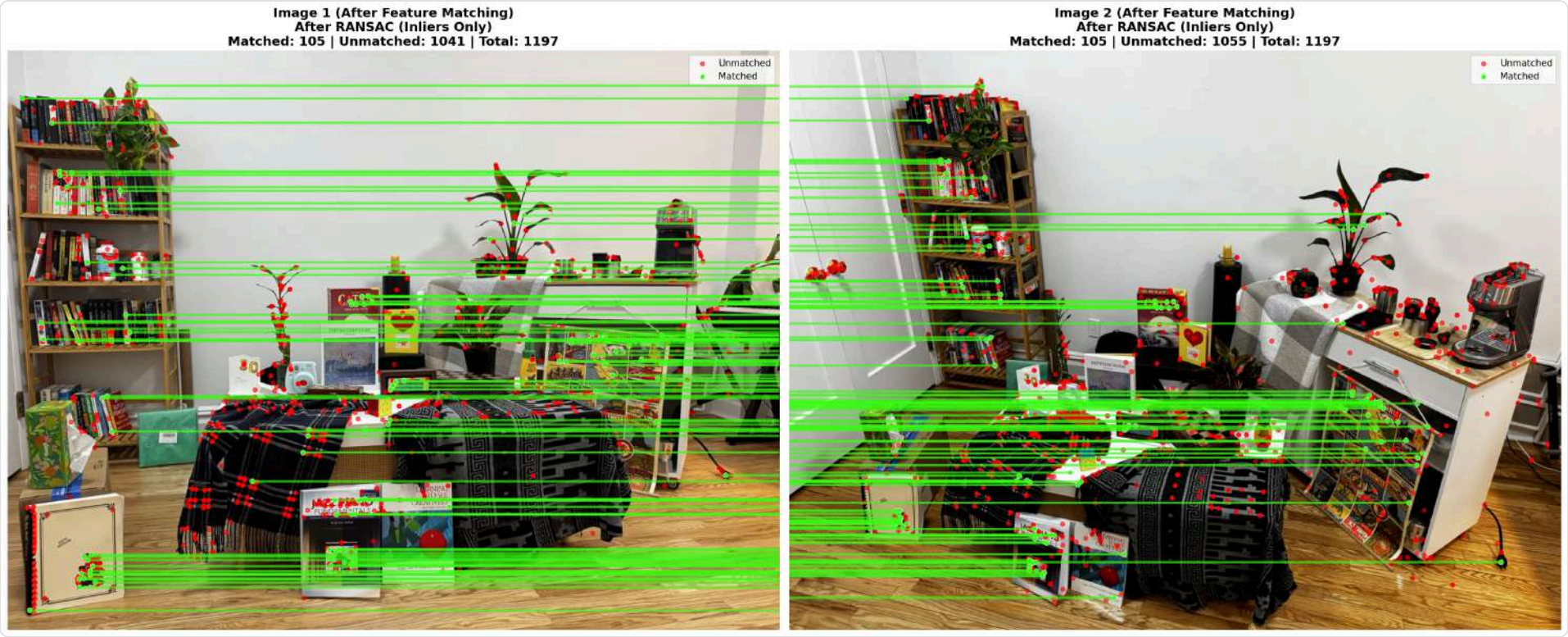
▼ Step 4: RANSAC Convergence



▼ Step 4: Correspondences After RANSAC (Inliers Only)

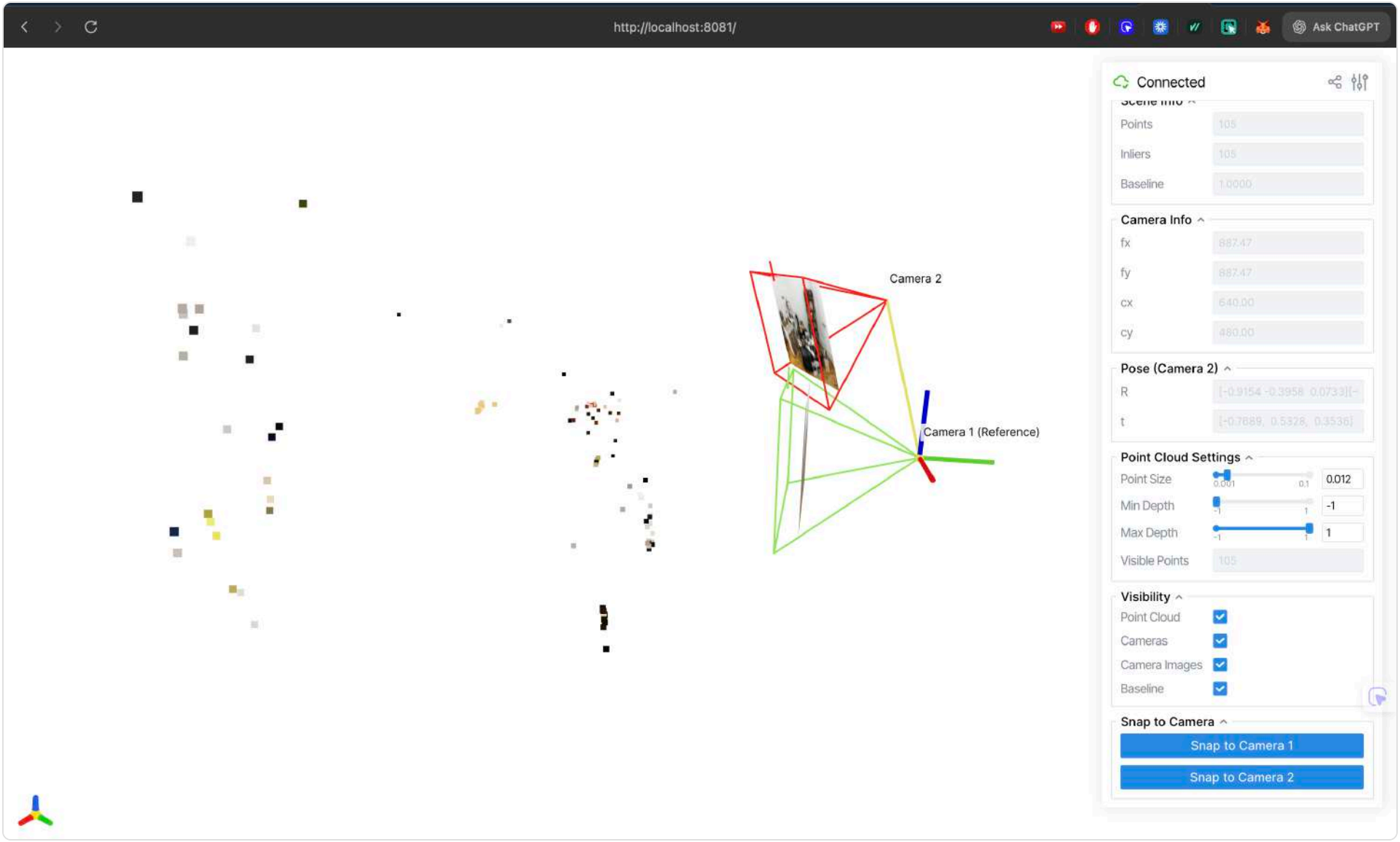


Alternate view (with patches):



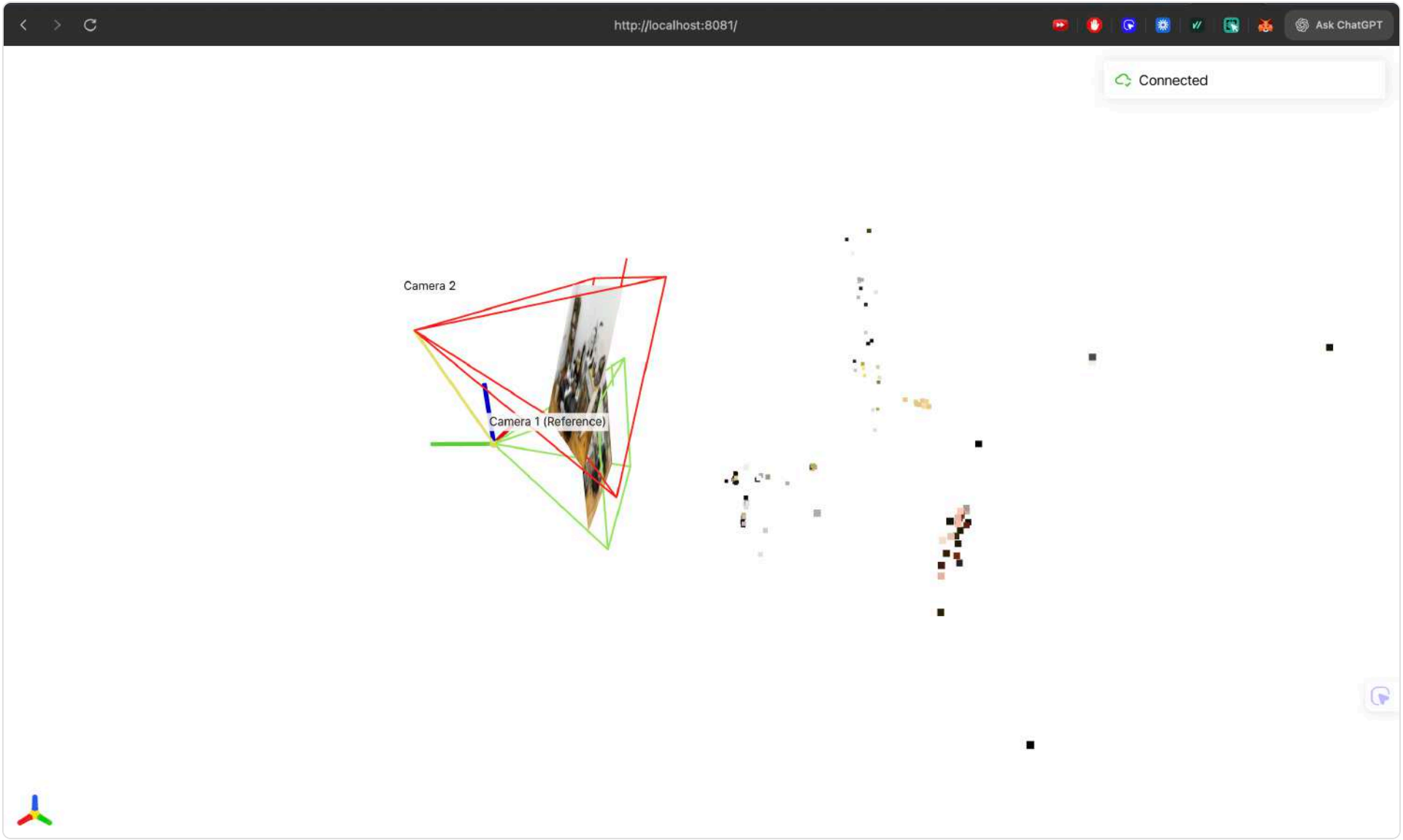
▼ Outputs: 3D Sparse Point Cloud (Viser)

▼ Landmark 1



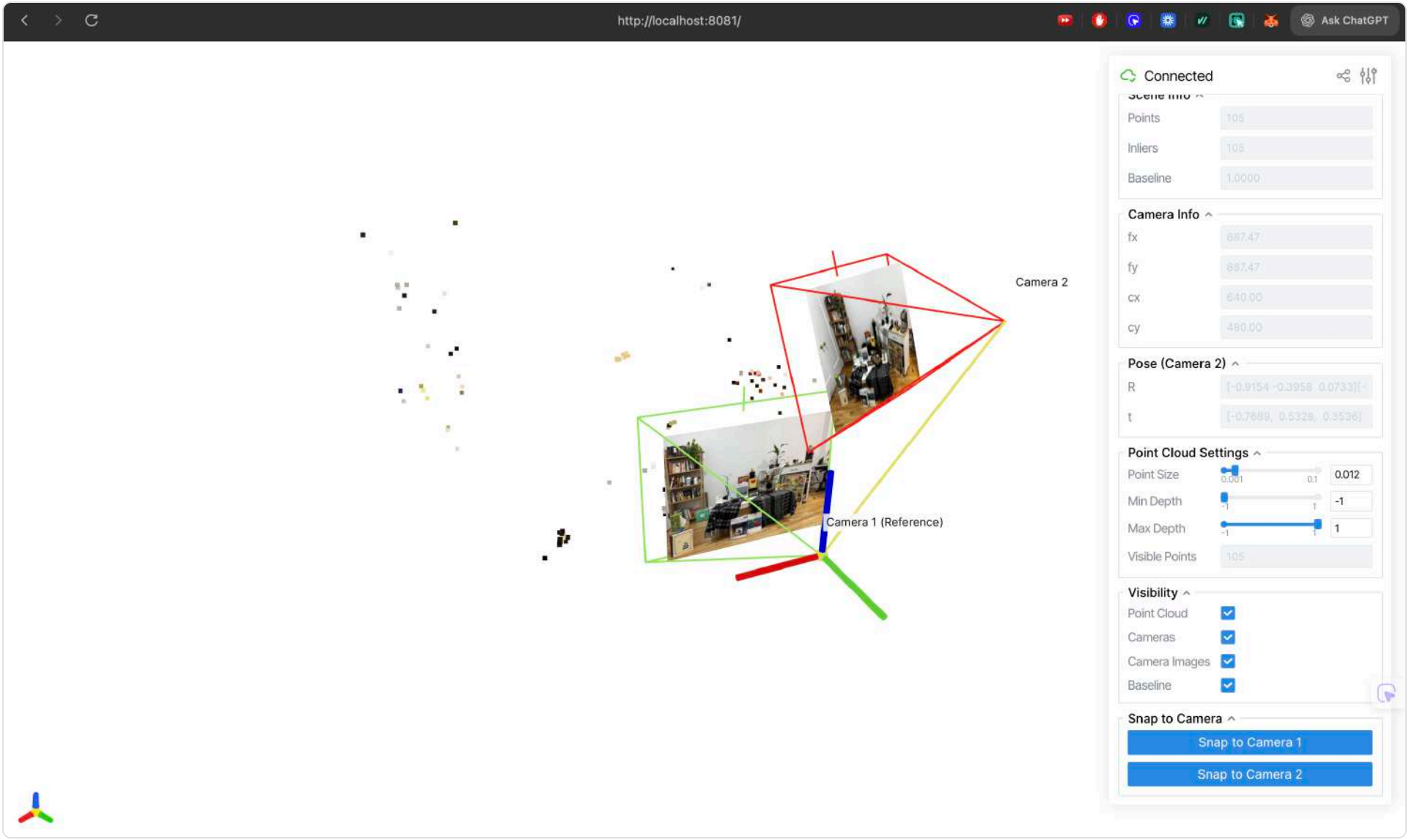
Cluster of reconstructed points matching the tall wooden bookshelf on the left side of the room.

▼ Landmark 2



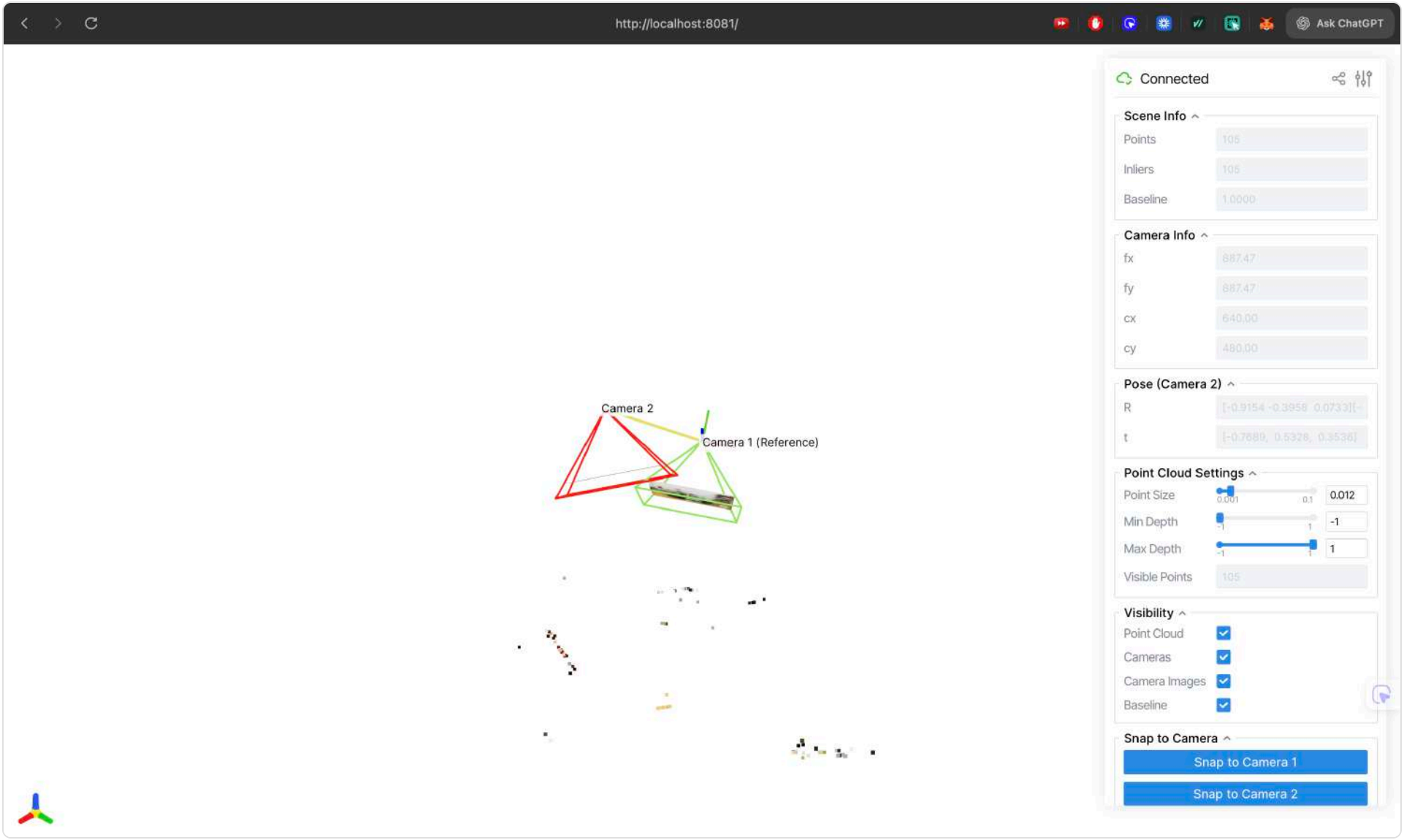
Reconstructed 3D geometry of the high-contrast blanket patterned over the bed in the center.

▼ Landmark 3



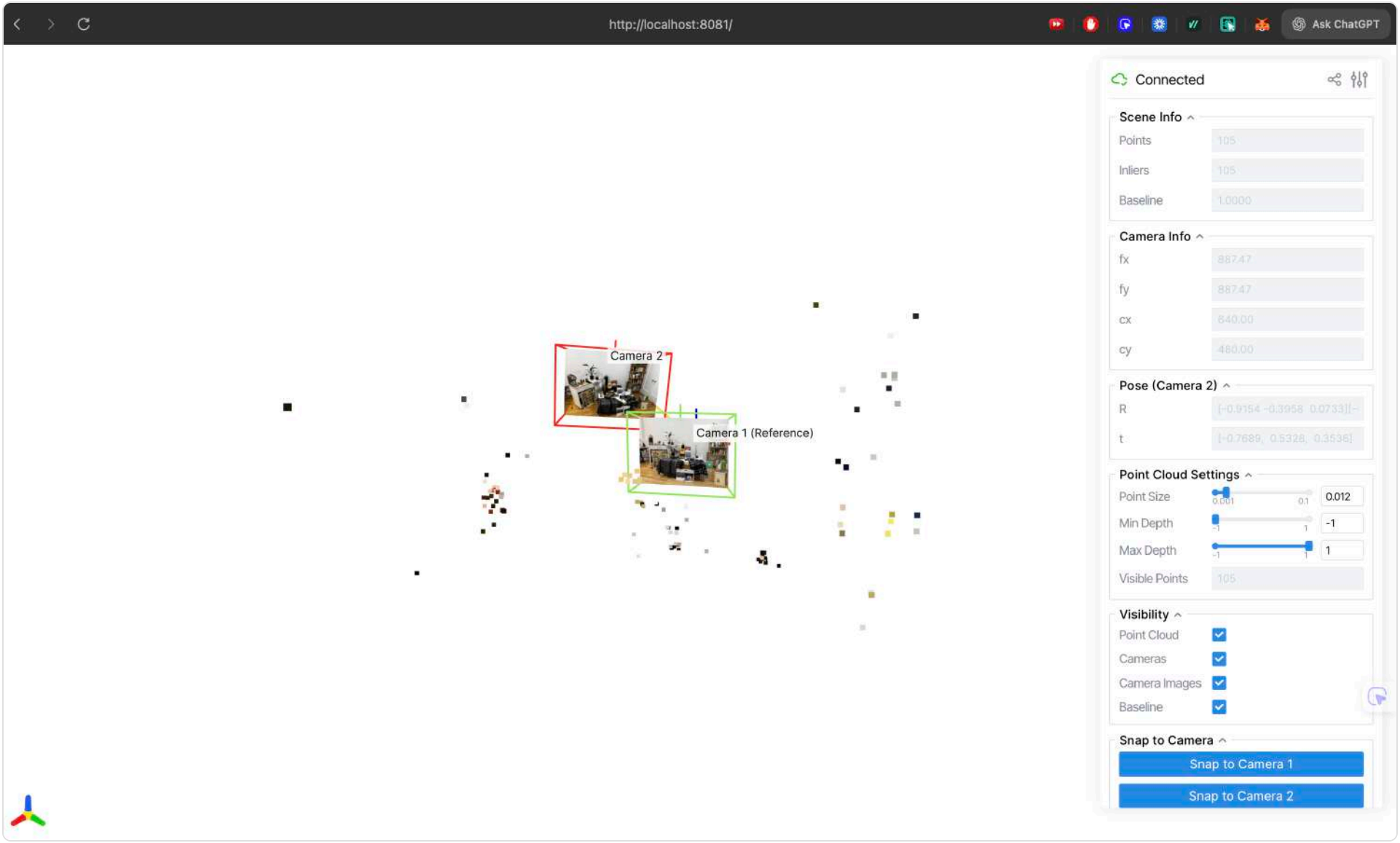
Dense points accurately capturing the structure of the small wooden drawers and items on the right side.

▼ Landmark 4



Points representing the edge of the large low table situated in front of the bed.

▼ Landmark 5



Wider perspective viewing the full back-wall structure including plants and decor above the bed.

Scene 2: Stone Doorway

▼ Visualizations — Pipeline Steps

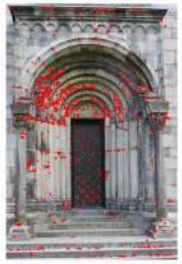
▼ Overview: Full Pipeline Grid

Complete Pose Estimation Pipeline

Image 1: Original

Image 2: Original

SIFT (1182)



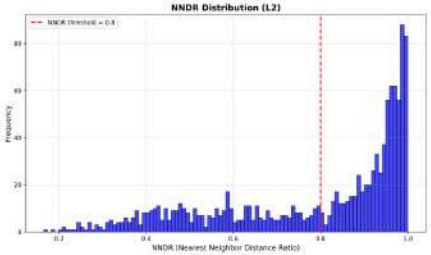
SIFT (1195)

SIFT (No ANMS) (1182)

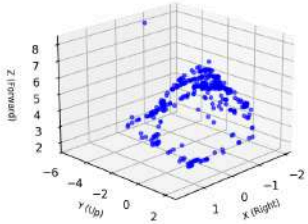
SIFT (No ANMS) (1195)

NNDR Top 5 Matches

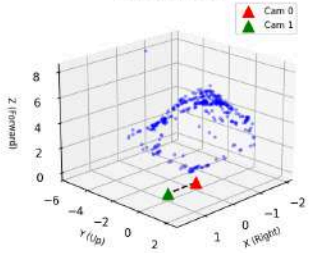
NNDR Ratio Histogram



3D Points (358)



Camera Poses



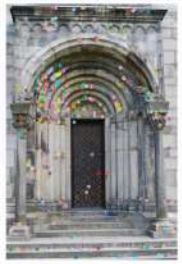
Matches Before RANSAC (381)

Matches Before RANSAC (381)

```
FEATURE DETECTION

SIFT:
  Image 1: 1182
  Image 2: 1195

ANMS Selected:
  Image 1: 1182
  Image 2: 1195
Config:
  Max features: 1200
  Contrast thresh: 0.04
  Edge thresh: 10

FEATURE MATCHING

Before RANSAC:
  Correspondences: 381

After RANSAC:
  Inliers: 361
  Inlier Ratio: 94.8%
Config:
  Metric: L2
  NNDR threshold: 0.8
```

```
POSE ESTIMATION

Translation:
  X: -0.9670
  Y: 0.0286
  Z: 0.2531

Baseline: 1.0000
RANSAC Config:
  Sample size (s): 8
  Epsilon: 0.0002
  Iterations: 2000

3D RECONSTRUCTION

Method: Sparse
Points: 358
Mean depth: 4.63
Triangulation Config:
  Method: Sparse features
  Max reproj error: 50.0px
  Depth range: [0.01, 10000]
```

Inliers After RANSAC (361)



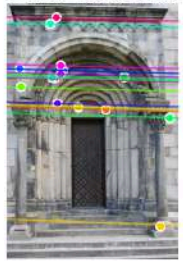
Inliers After RANSAC (361)

Image 1 - Feature Points

Image 2 - Epipolar Lines

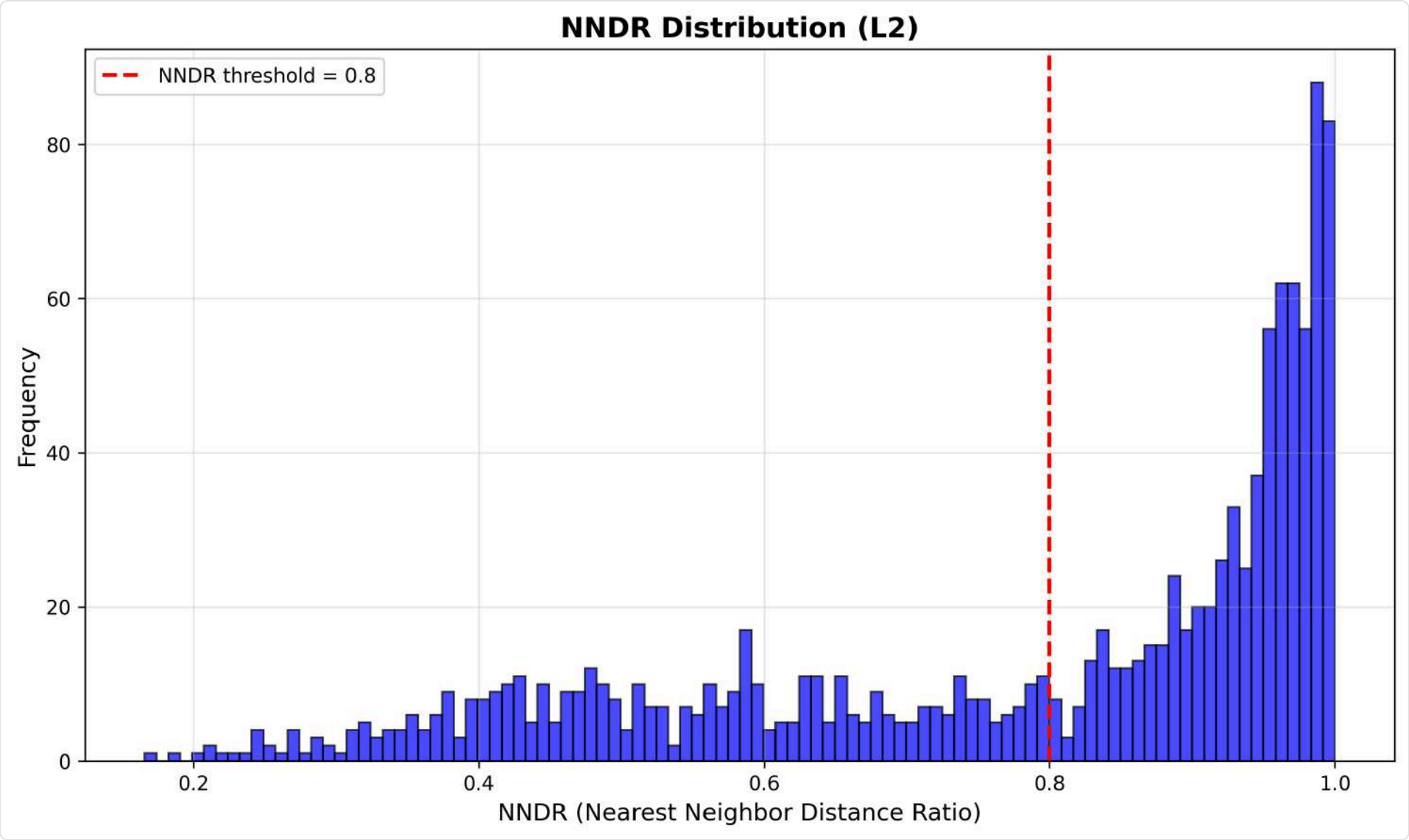
Image 2 - Feature Points

Image 1 - Epipolar Lines

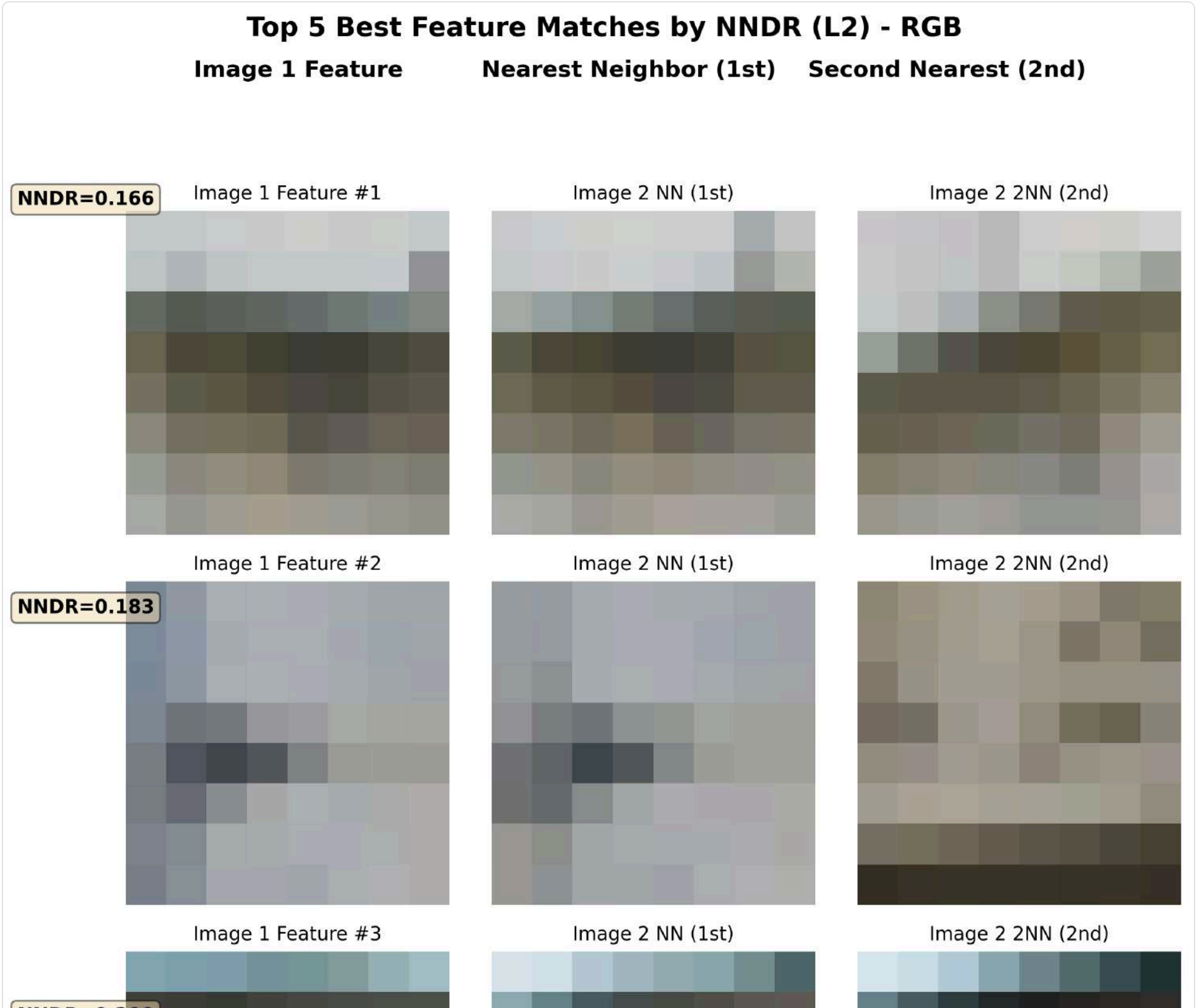


▼ Step 2: SIFT Features Detected

▼ Step 3: NNDR Histogram



▼ Step 3: Top 5 Features Matched (Ranked by NNDR)


Top 5 Best Feature Matches by NNDR (L2) - RGB
Image 1 Feature
Nearest Neighbor (1st)
Second Nearest (2nd)
NNDR=0.166
Image 1 Feature #1
Image 2 NN (1st)
Image 2 2NN (2nd)
NNDR=0.183
Image 1 Feature #2
Image 2 NN (1st)
Image 2 2NN (2nd)
Image 1 Feature #3
Image 2 NN (1st)
Image 2 2NN (2nd)

NNDR=0.200
Image 1 Feature #4
Image 2 NN (1st)
Image 2 2NN (2nd)
NNDR=0.209
Image 1 Feature #5
Image 2 NN (1st)
Image 2 2NN (2nd)
NNDR=0.210

▼ Step 3: Correspondences Before RANSAC

Image 1 - Before RANSAC (n=381) (381 correspondences)

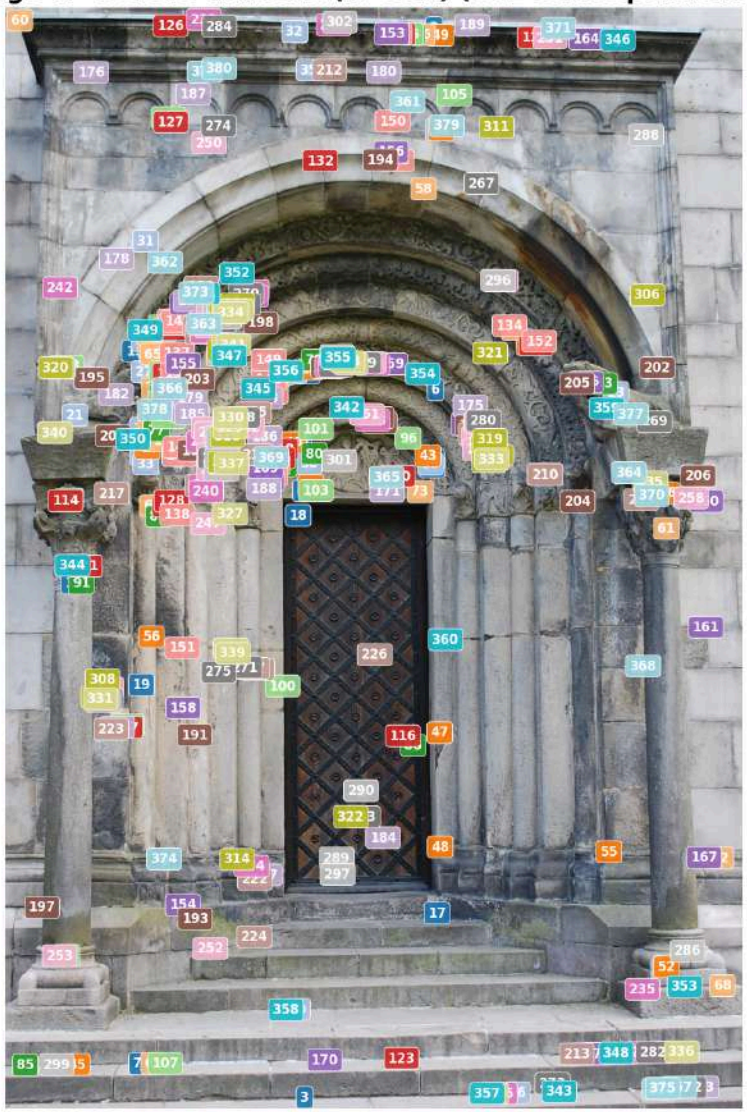
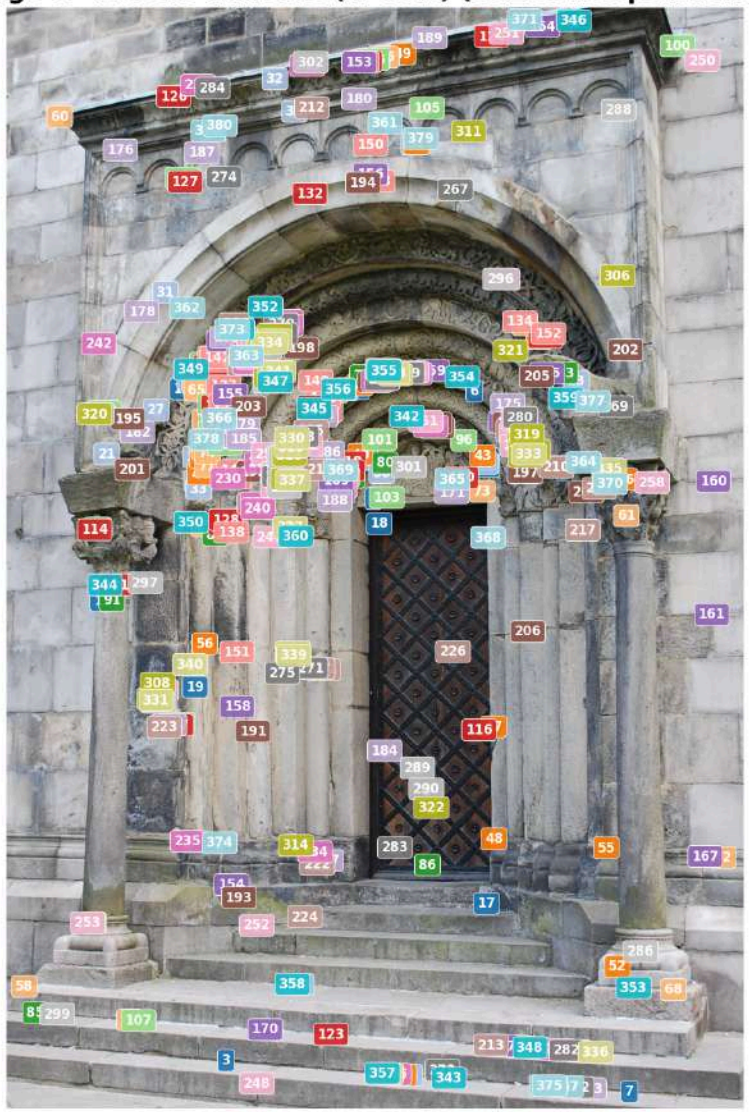
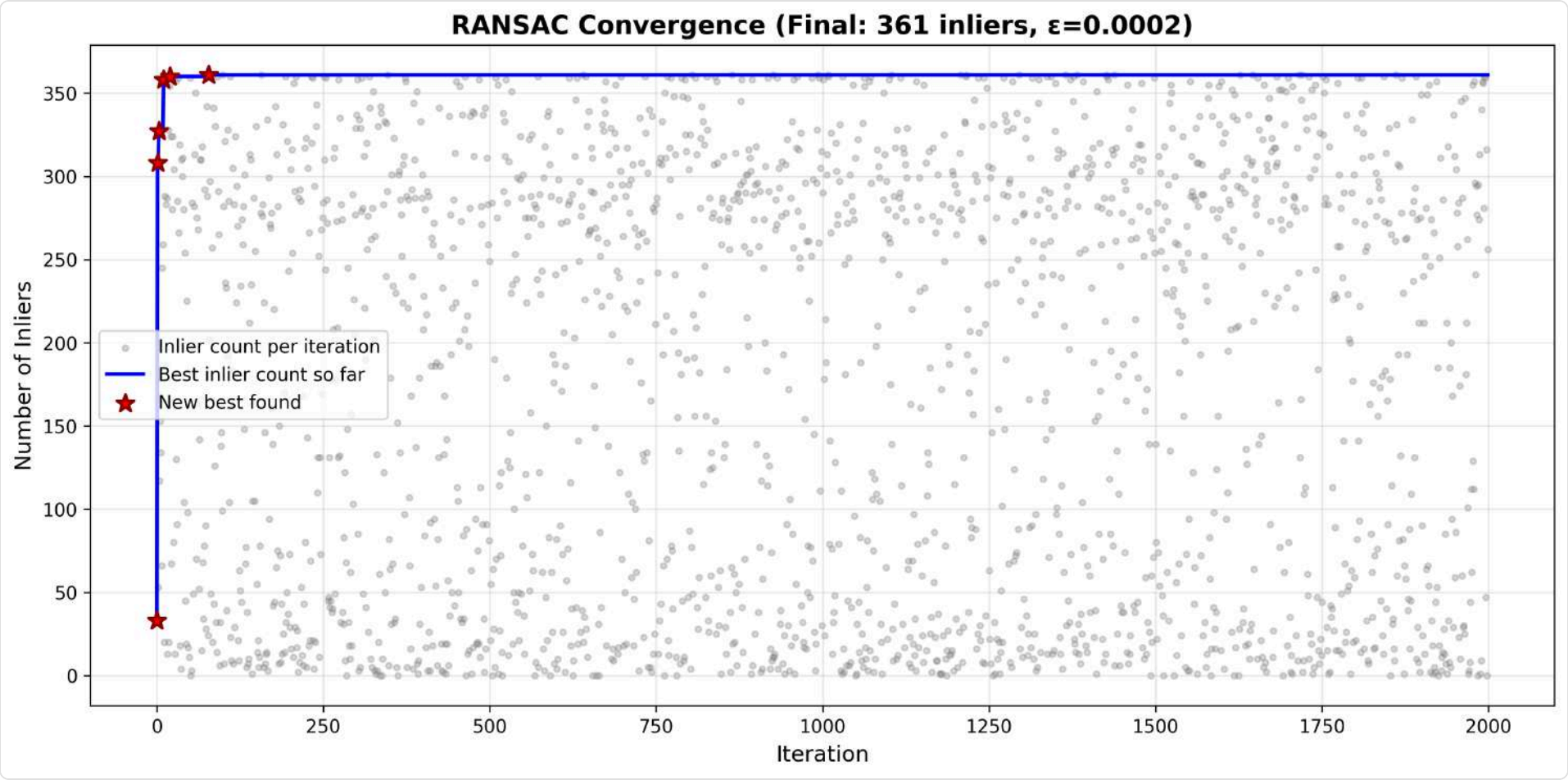


Image 2 - Before RANSAC (n=381) (381 correspondences)



▼ Step 4: RANSAC Convergence



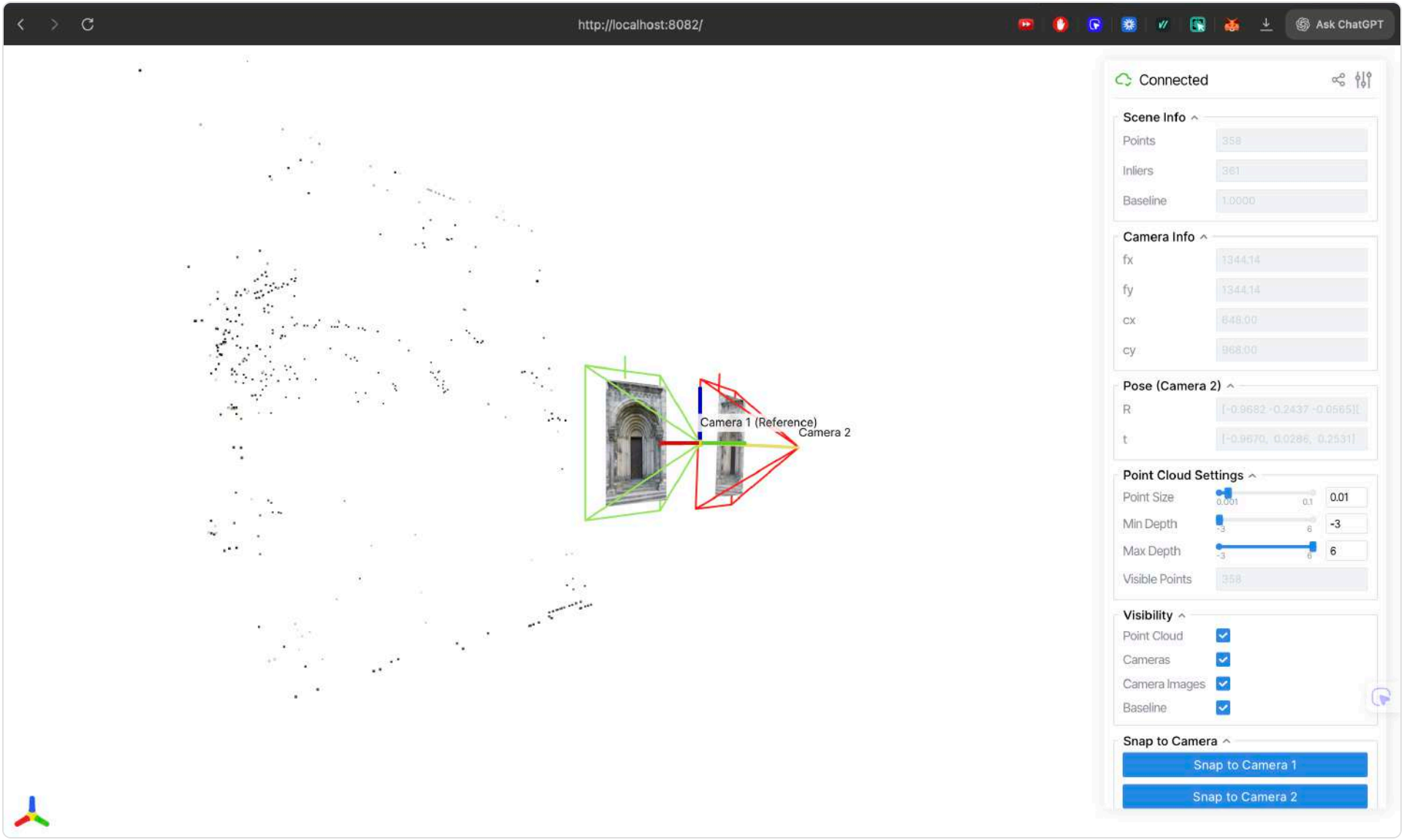
▼ Step 4: Correspondences After RANSAC (Inliers Only)

▼ Outputs: 3D Sparse Point Cloud (Viser)

▼ Landmark 1

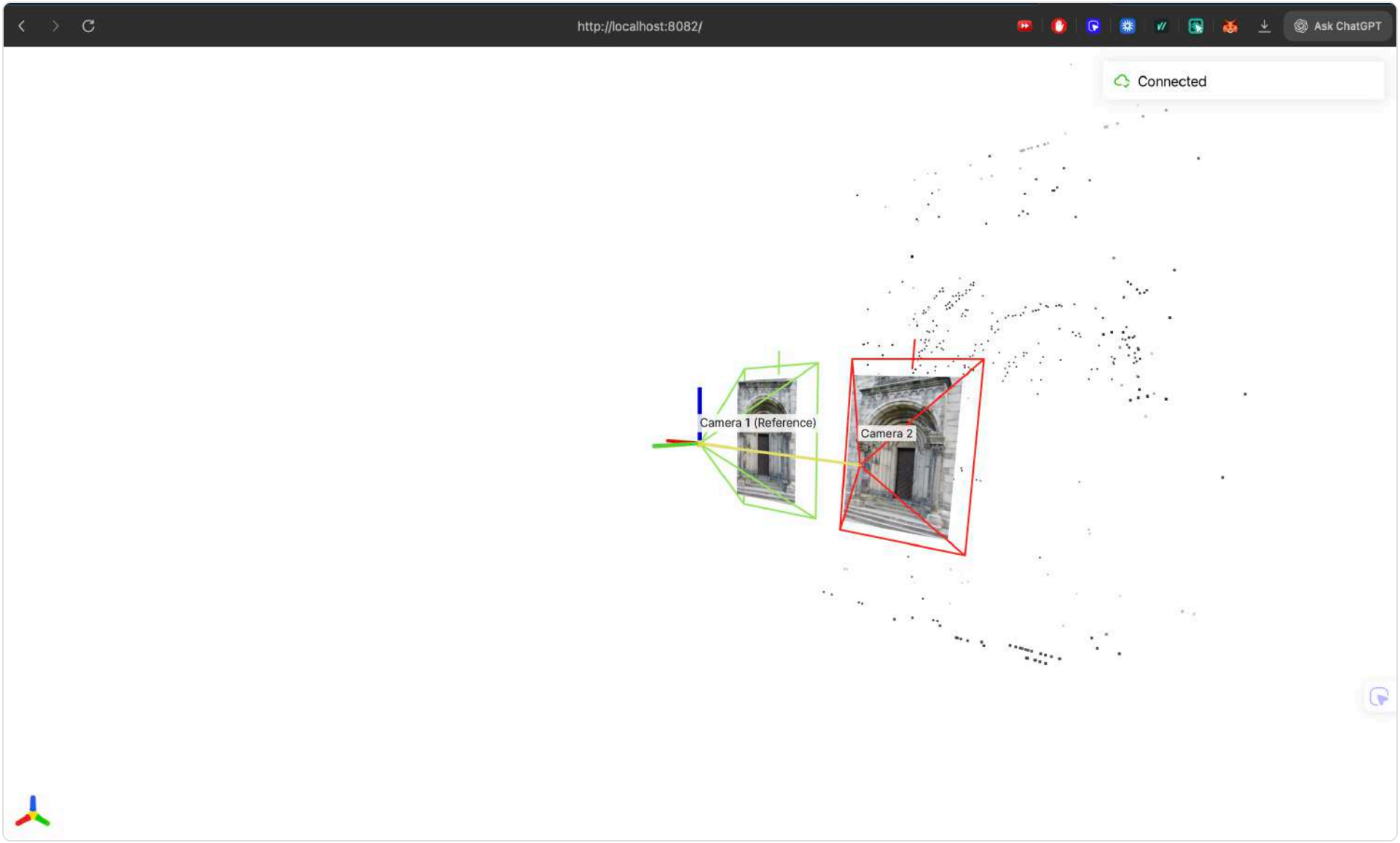
Wide contextual view showing both camera poses and the central dense point cluster representing the stone archway.

▼ Landmark 2



Direct head-on view from Camera 1's perspective indicating accurately matched points along the stone pillars and steps.

▼ Landmark 3



Top-down alternate angle showing the depth reconstruction of the stairs retreating into the doorway.